



SPS2_205: Computed Tomography in the evaluation of acute injuries of the anterior eye segment

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The authors have no disclosures

PURPOSE

- To evaluate the diagnostic performance of CT in acute anterior segment ocular injuries as an adjunctive diagnostic modality.

Standard diagnostic techniques:

- Slit lamp ophthalmologic examination
- Ocular B-scan sonography
- Constrained by technical limitations



MATERIALS AND METHODS

- Retrospective design
- 122 patients, presented to the ED (from January 2014 to April 2016) with recent direct trauma to the anterior segment of the eye
- De-identified multiplanar thin-slice CT images were reviewed by two subspecialty board-certified neuroradiologists:
 - Anterior segment rupture
 - Hyphema
 - lens, ciliary body, and lacrimal gland injury
- CT findings were compared to slit lamp, B-scan sonography, and/or operative data as the gold standard

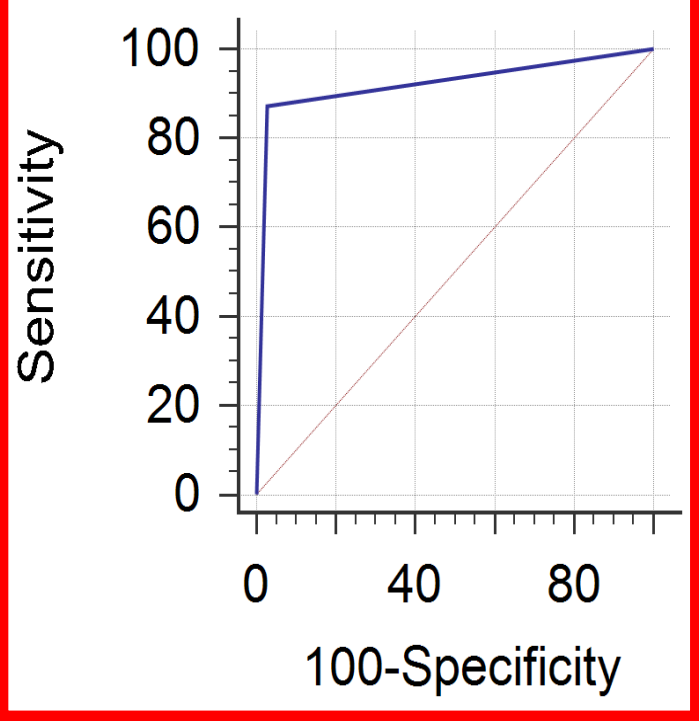
RESULTS

Table 1: Numbers of patients according to the site of injury

Site of injury	n = 122
Preseptal compartment	91
Lacrimal system	7
Foreign body (corneal or intra-ocular)	19
Corneal injury (partial or full thickness)	64
Hyphema	53 (64 including microhyphema)
Lens dislocation/subluxation	25
Iris injury/prolapse	24
Rupture anterior globe	47

RESULTS

Table 2: Diagnostic performance of our neuroradiologists' CT diagnosis of anterior globe rupture/perforation

Globe rupture/ perforation	Sensitivity	Specificity		AUC	Inter-rater agreement (kappa)
Rater 1	87.23% CI = 74.3-95.2%	94.67% CI = 86.9-98.5%		0.910 P < 0.0001	0.876 CI = 0.787- 0.965
Rater 2	85.11% CI = 71.7-93.8%	94.67% CI = 86.9-98.5%		0.899 P < 0.0001	
Consensus rating	87.23% CI = 74.3-95.2%	97.33% CI = 90.7-99.7%		0.923 P < 0.0001	

RESULTS

Table 3: Diagnostic performance per consensus rating by our neuroradiologists' CT interpretation of different traumatic findings. CI = 95% confidence interval

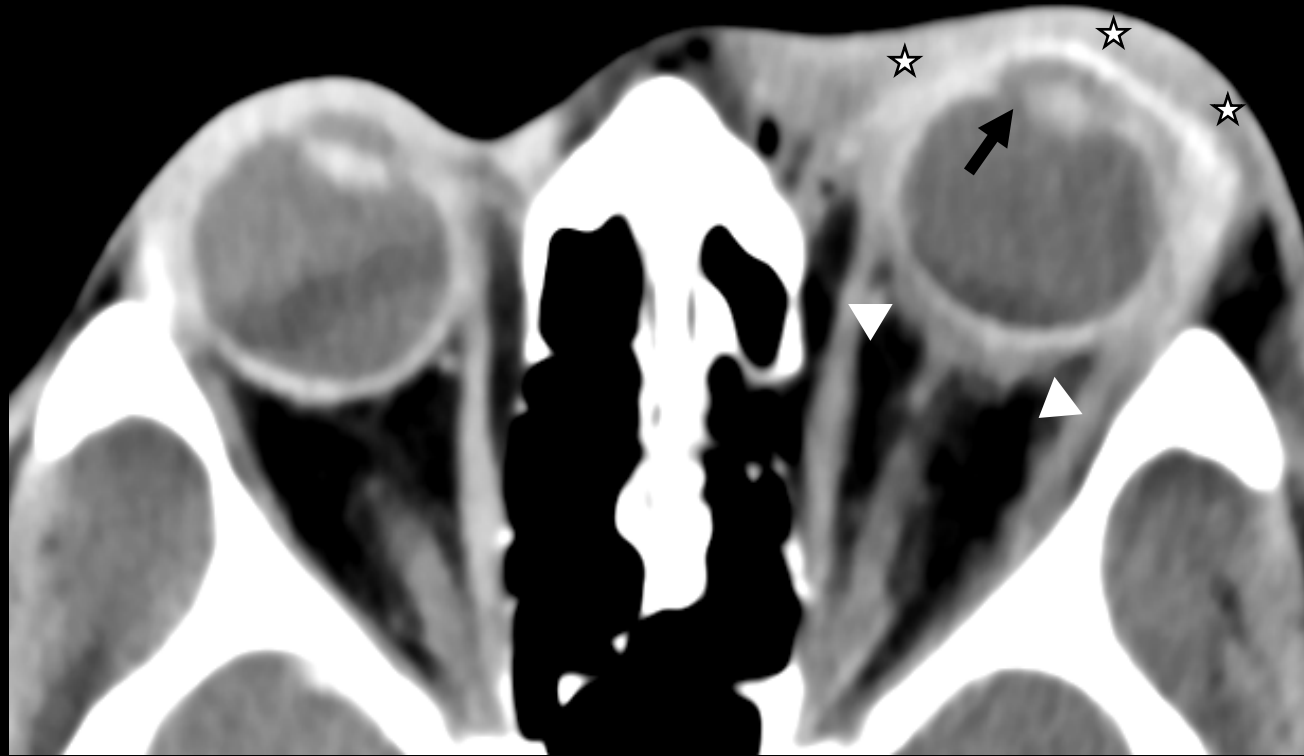
Traumatic finding	Sensitivity	Specificity	Area under ROC curve (AUC)	Inter-rater agreement (kappa)
Preseptal swelling	84.62% CI = 75.54-91.33%	80.65% CI = 62.53-92.55%	0.83 P < 0.0001	0.472 CI = 0.302-0.643
Foreign body	<u>68.42%</u> CI = 43.4-87.4%	100% CI = 96.5-100.0%	0.842 P < 0.0001	<u>0.914</u> CI = 0.796-1.000
Corneal injury	<u>50%</u> CI = 37.2-62.8%	93.1% CI = 83.3-98.1%	0.716 P < 0.0001	0.710 CI = 0.578-0.842
Anterior hyphema	77.36% CI = 63.8-87.7%	88.41% CI = 78.4-94.9%	0.829 P < 0.0001	0.781 CI = 0.670-0.891
Shallow anterior chamber	89.19% CI = 74.6-97.0%	87.06% CI = 78.0-93.4%	0.881 P < 0.0001	0.766 CI = 0.653-0.878
Lens dislocation/subluxation	88% CI = 68.8-97.5%	90.72% CI = 83.1-95.7%	0.894 P < 0.0001	0.751 CI = 0.619-0.884

RESULTS

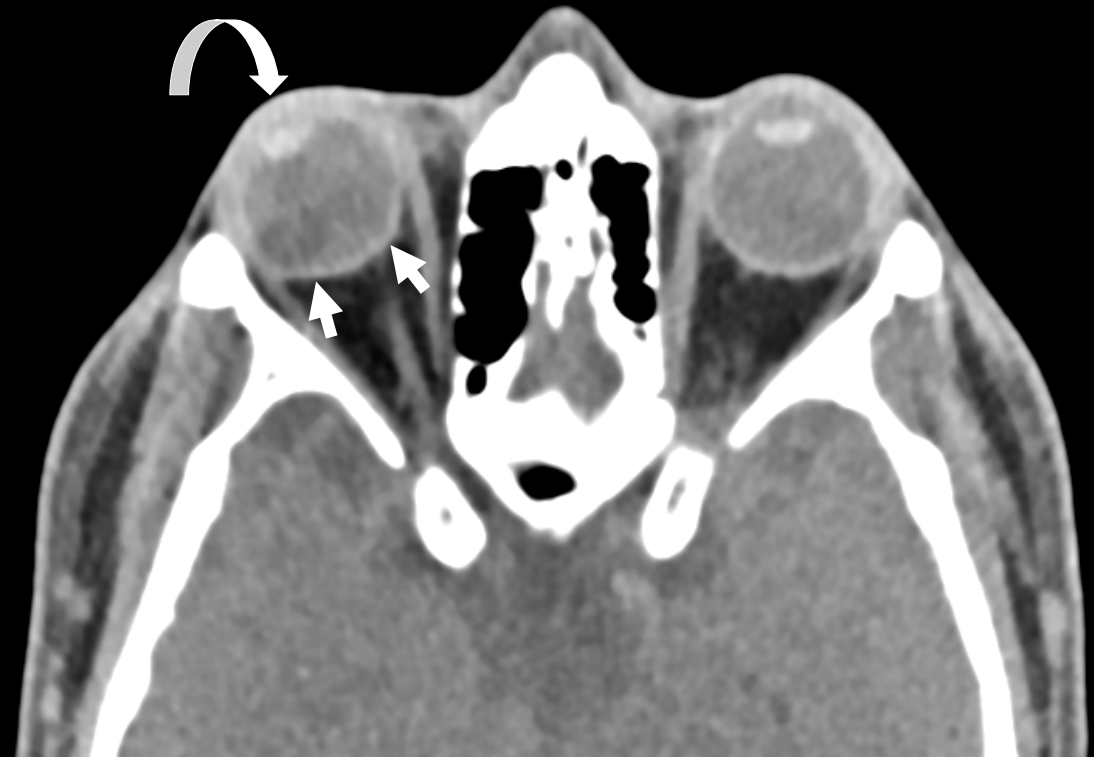
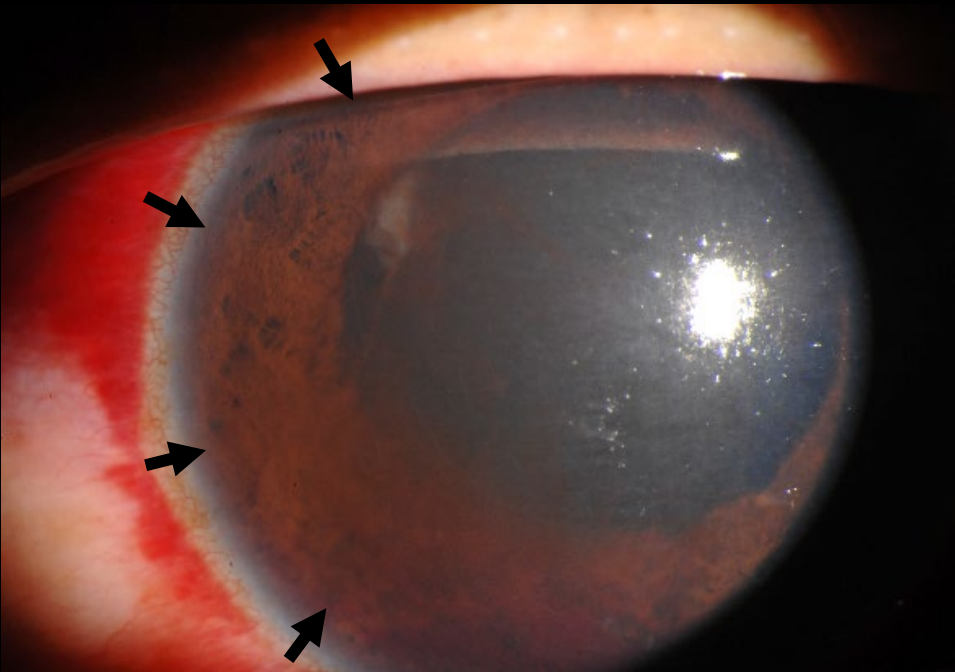
Table 4: Chi-squared analysis of association and Odds ratio for abnormal depth and density of the anterior chamber as risk predictors of anterior globe rupture/perforation. CI = 95% confidence interval.

	Chi-squared	Odds ratio	CI	P value
Increased density of anterior chamber	69.91	57.23	18.53-176.74	< 0.0001
Shallow anterior chamber	72.37	68.25	20.89-223.0	< 0.0001

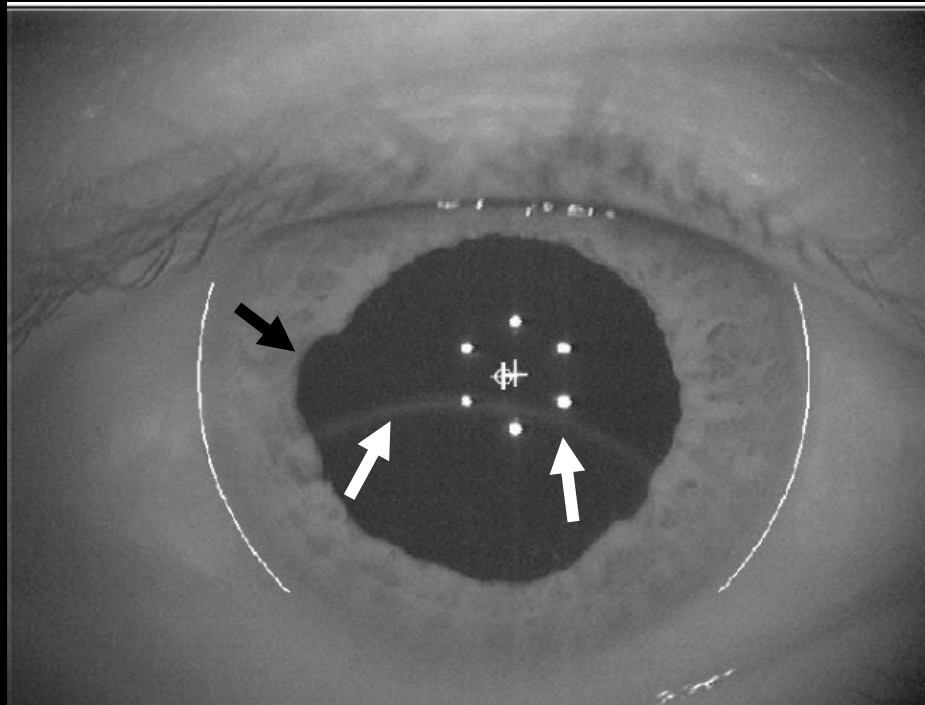
CASE 1: Left lenticular anterior subluxation into the AC due to torn medial suspensory ligament (arrow) giving a false appearance of a narrowed AC in a patient with blunt left eye trauma. Note also preseptal ecchymosis (stars) and the thin retrobulbar hematoma (arrowheads).



CASE 2: A patient with ruptured anterior right globe and hyphema. Slit lamp examination (left image) showing grade IV hyphema seen as blood filling the whole volume of the anterior chamber (black arrows) with CT correlate of the same patient (right image) demonstrating high density within the right anterior chamber that appears abnormally shallow (curved arrow). The globe outline is also deformed with relative loss of volume (straight white arrows).



CASE 3: Posterior lens dislocation in a patient with history of trauma. Left image shows LENS-Master ophthalmoscopic visualization of dislocated lens (white arrows) with CT correlate of the same patient (right image) demonstrating posterior lens dislocation. Iris focal damage is also noted (black arrow in left image).



CASE 4: A tiny (2-mm) hypodense (40-HU) wooden intra-corneal foreign body (arrow) that was missed by our two raters in a patient who sustained left open globe injury while he was cutting weeds. Note also marked narrowing of left anterior chamber due to anterior globe perforation.



CONCLUSION

- CT can detect **anterior rupture globe** with a diagnostic accuracy of 94%.
- Subtle ocular findings on CT, including SHALLOW ANTERIOR CHAMBER, can provide valuable information to the ophthalmologist for the diagnosis of occult anterior segment rupture.
- Our neuroradiologists were also able to diagnose **lens dislocation, anterior hyphema, and metallic foreign bodies** satisfactorily.
- Understanding the diagnostic capability of CT will help establish its CRITICAL role as an ADJUNCT modality in addition to clinical ophthalmologic methods of assessment in the emergency workup of patients with trauma to the anterior eye segment.

Thank you

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